# Exercise Sheet 5

Throughout, $\int_G f(g)\,dg$ is the normalised Haar integral of Lecture 5. Inner products are antilinear in the first slot. Exercises 5.1–5.6 are the core work. Exercise 5.7 is an optional extension.

**Exercise 5.1. The circle and its finite subgroups.** Let $\mathrm{U}(1) = \{z \in \mathbb{C} : |z| = 1\}$ with

$$\int_{\mathrm{U}(1)} f(z)\,dz = \frac{1}{2\pi}\int_0^{2\pi} f(e^{i\theta})\,d\theta.$$

Fix $N \geq 2$, put $\zeta = e^{2\pi i/N}$ and $\mu_N = \{\zeta^k : 0 \leq k < N\}$. For a function $f$ on $\mathrm{U}(1)$, let

$$A_N(f) = \frac{1}{N}\sum_{k=0}^{N-1} f(\zeta^k).$$

(a) For $n \in \mathbb{Z}$, compute $\int_{\mathrm{U}(1)} z^n\,dz$ and $A_N(z^n)$.

(b) Let $f(z) = \sum_{|n|\leq M} c_n z^n$ with $M < N$. Show that $A_N(f) = \int_{\mathrm{U}(1)} f(z)\,dz$. Give an $f$ with $M = N$ for which the two values differ.

(c) Let $\pi(z) = \mathrm{diag}(z^N, z)$ on $\mathbb{C}^2$. Compute $P = \int_{\mathrm{U}(1)} \pi(z)\,dz$ and $P_N = \frac{1}{N}\sum_k \pi(\zeta^k)$. Identify their images.

**Exercise 5.2. Moments on SU(2).** Write

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix} \in \mathrm{SU}(2), \qquad a = \cos\eta\, e^{i\xi_1}, \quad b = \sin\eta\, e^{i\xi_2},$$

with $0 \leq \eta \leq \pi/2$ and $0 \leq \xi_1, \xi_2 \leq 2\pi$. Use the Haar integral

$$\int_{\mathrm{SU}(2)} f(g)\,dg = \frac{1}{2\pi^2}\int_0^{2\pi}\int_0^{2\pi}\int_0^{\pi/2} f\,\cos\eta\sin\eta\,d\eta\,d\xi_1\,d\xi_2.$$

Let $p, q, r, s \geq 0$ be integers.

(a) Show that $\int_{\mathrm{SU}(2)} a^p \bar{a}^q b^r \bar{b}^s\,dg = 0$ unless $p = q$ and $r = s$.

(b) Show that

$$\int_{\mathrm{SU}(2)} |a|^{2p}|b|^{2r}\,dg = \frac{p!\,r!}{(p+r+1)!}.$$

(c) Deduce the integrals of $|a|^2$, $|a|^4$, $|b|^4$ and $|a|^2|b|^2$. Check them against $|a|^2 + |b|^2 = 1$.

**Exercise 5.3. Coefficient norms in $V_2$.** Let $g \in \mathrm{SU}(2)$ be written as

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad |a|^2 + |b|^2 = 1.$$

You may use these facts. A monomial $a^p\bar{a}^q b^r \bar{b}^s$ has Haar integral 0 unless $p = q$ and $r = s$. Moreover $\int |a|^2 = \int |b|^2 = \frac{1}{2}$, $\int |a|^4 = \int |b|^4 = \frac{1}{3}$ and $\int |a|^2|b|^2 = \frac{1}{6}$. Let $e_1, e_2$ be the standard basis of $\mathbb{C}^2$, and write $e_{jk} = e_j \otimes e_k$ in $\mathbb{C}^2 \otimes \mathbb{C}^2$ with the tensor-product inner product. Let $V_2$ be spanned by

$$f_0 = e_{11}, \qquad f_1 = \tfrac{1}{\sqrt{2}}(e_{12} + e_{21}), \qquad f_2 = e_{22},$$

with $\pi(g) = g \otimes g$ restricted to $V_2$. You may assume that $V_2$ is invariant and irreducible of dimension 3. Put $D_{ij}(g) = \langle f_i, \pi(g) f_j\rangle$.

(a) Show that $f_0, f_1, f_2$ are orthonormal. Compute $\pi(g)f_0$ and $\pi(g)f_1$ in this basis.
(b) Write down the first two columns of $D(g)$.
(c) Verify $\int_{\mathrm{SU}(2)} |D_{ij}|^2 \, dg = \tfrac{1}{3}$ for $j = 0, 1$ and $i = 0, 1, 2$.
(d) Verify $\int \overline{D_{00}}\, D_{11} \, dg = 0$ and $\int D_{11} \, dg = 0$. Say which case of Schur orthogonality each identity illustrates.

**Exercise 5.4. The invariant-vector projector on matrices.** Let $\mathrm{SU}(2)$ act on $M_2(\mathbb{C})$ by $\mathrm{Ad}(g)X = gXg^{-1}$. Use the inner product $\langle X, Y\rangle = \operatorname{tr}(X^*Y)$, and let $E_{jk}$ be the matrix units. Write

$$g = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad g^{-1} = g^* = \begin{pmatrix} \bar{a} & \bar{b} \\ -b & a \end{pmatrix}.$$

You may use the orthogonality relation for $V_1 = \mathbb{C}^2$, namely $\int \overline{g_{kl}}\, g_{ij} \, dg = \tfrac{1}{2}\delta_{ik}\delta_{jl}$. You may also use $\int |a|^2 = \int |b|^2 = \tfrac{1}{2}$ and the rule that $a^p \bar{a}^q b^r \bar{b}^s$ integrates to 0 unless $p = q$ and $r = s$.

(a) Show that the $E_{jk}$ are orthonormal and that each $\mathrm{Ad}(g)$ is unitary.
(b) Compute $P(X) = \int_{\mathrm{SU}(2)} gXg^{-1} \, dg$ for every $X$.
(c) Check (b) directly for $X = E_{11}$ and $X = E_{12}$.
(d) Identify the invariant matrices. Confirm that $P$ is the orthogonal projection onto them.

**Exercise 5.5. Averaging a non-invariant inner product.** Let $\mathrm{U}(1)$ carry the Haar integral of Exercise 5.1, so $\int z^n \, dz = \delta_{n0}$. Put

$$S = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \pi(z) = S \begin{pmatrix} z & 0 \\ 0 & \bar{z} \end{pmatrix} S^{-1} \quad \text{on } \mathbb{C}^2,$$

with the standard inner product $\langle v, w\rangle_0 = v^*w$.

(a) Compute $\pi(z)$. Show that $\mathbb{C}e_1$ is invariant but its standard complement $\mathbb{C}e_2$ is not.
(b) Compute $H = \int \pi(z)^*\pi(z) \, dz$, so that the averaged inner product is $\langle v, w\rangle = v^*Hw$. Show that $H$ is positive definite.
(c) Verify $\pi(z)^*H\pi(z) = H$ by multiplying matrices.
(d) Find the $H$-orthogonal complement of $\mathbb{C}e_1$ and check that it is invariant.
(e) Which property of the Haar integral guarantees (c) without computation?

**Exercise 5.6. Inversion invariance and characters.** Let $G$ be a compact Lie group. You may use left and right invariance, positivity and normalisation of the Haar integral. You may also use the cited uniqueness theorem: a linear functional $J$ on $C(G)$ with $J(f) \geq 0$ for $f \geq 0$, $J(1) = 1$ and $J(f \circ L_h) = J(f)$ for all $h$ is the Haar integral.

(a) Let $J(f) = \int_G f(g^{-1}) \, dg$. Show that $J$ satisfies the hypotheses of the uniqueness theorem. Conclude that $\int_G f(g^{-1}) \, dg = \int_G f(g) \, dg$.
(b) Let $\pi$ be a unitary representation with character $\chi_\pi(g) = \operatorname{tr}\pi(g)$. Show that $\chi_\pi(g^{-1}) = \overline{\chi_\pi(g)}$.
(c) Let $\pi, \sigma$ be irreducible unitary representations. Using the character orthonormality of Lecture 5, show that $\int_G \chi_\pi(g)\chi_\sigma(g^{-1}) \, dg$ equals 1 if $\pi \cong \sigma$ and 0 otherwise.

**Exercise 5.7. Optional extension: where compactness enters.** Let

$$G = \left\{ g(\alpha,\beta) = \begin{pmatrix} \alpha & \beta \\ 0 & 1 \end{pmatrix} : \alpha > 0,\ \beta \in \mathbb{R} \right\}.$$

This is a noncompact Lie group with global chart $(\alpha, \beta)$. Its Lie algebra has basis $X = E_{11}$ and $Y = E_{12}$. A continuous density $\delta$ on $G$ is given by its coefficient $\rho(\alpha, \beta)$ in this chart. If a diffeomorphism $F$ is written in coordinates as $u \mapsto F(u)$, then $F^*\delta$ has coefficient $u \mapsto \rho(F(u))\,|\det DF(u)|$.

(a) Show that $g(\alpha,\beta)\,g(\alpha',\beta') = g(\alpha\alpha', \alpha\beta' + \beta)$. Compute $\mathrm{Ad}(g)X$, $\mathrm{Ad}(g)Y$ and $j(g) = |\det \mathrm{Ad}(g)|$.

(b) Show that $\rho_L = \alpha^{-2}$ is invariant under all $L_h$, and $\rho_R = \alpha^{-1}$ is invariant under all $R_h$.

(c) Show that $R_h^*$ multiplies $\rho_L$ by $j(h)^{-1}$. Which step of the right-invariance proof in Lecture 5 fails here?